Lava-Android-1.2.0-1020 — Release Notes (SP-3a)

> **Status:** Feature-complete (HEAD `b899474` at the start of the documentation-polish pass; subsequent commits are doc-only). Tagging is **BLOCKED** pending operator real-device attestation per Tasks 5.22 / 5.25 / 4.20 — see Operator verification checklist below.
>
> **Audience:** Lava operators preparing to cut the 1.2.0 tag, plus downstream users wanting a one-page summary of what landed.

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1. What shipped

The SP-3a "Multi-Tracker SDK Foundation" arc replaces the rutracker-only Lava client with a multi-tracker SDK. Lava-Android-1.2.0-1020 ships with two trackers — **RuTracker** (existing, decoupled in Phase 2) and **RuTor** (new, added in Phase 3) — and the scaffolding to add a third without touching the app module.

User-visible features:

- **Tracker selector in Settings → Trackers.** RuTracker (the existing default) and RuTor (new) appear side by side; tapping either sets it active. The active tracker drives every search / browse / topic / download invocation.
- **Per-tracker mirror lists.** Bundled defaults plus user-added customs. Each row carries a colored health dot (HEALTHY / DEGRADED / UNHEALTHY / UNKNOWN).
- **Add custom mirror dialog.** Operators can extend the bundled set per tracker; entries persist in the `tracker_mirror_user` Room table.
- **Cross-tracker fallback modal.** When all mirrors of the active tracker hit UNHEALTHY, the SDK prompts the user with "All RuTracker mirrors are unhealthy. Try RuTor?". Accept → re-issues the query on the alt tracker. Dismiss → explicit failure UI (snackbar). **No silent fallback** is possible.
- **15-minute health probe.** A WorkManager `PeriodicWorkRequest` HEAD-probes every registered mirror every 15 minutes and updates the persisted health state.
- **Anonymous-by-default RuTor flow.** Searching, browsing, viewing topics, and downloading torrents work without a login. Login is invoked only when the user attempts an authenticated operation (post a comment, in 1.2.0).

Internal SDK shape:

- A new `:core:tracker:*` module family (api, registry, mirror, client, rutracker, rutor, testing).
- A Hilt-injected orchestrator (`LavaTrackerSdk`) that feature ViewModels consume.

- A pluggable per-tracker plugin shape — adding a third tracker is a documented 7-step recipe in `docs/sdk-developer-guide.md`.
- Generic primitives (`MirrorUrl`, `Protocol`, `Mirror` state machine, in-memory registry, mirror-config store, test scaffolding) extracted to a `vasic-digital/Tracker-SDK` submodule mounted at `submodules/tracker_sdk/` per the Decoupled Reusable Architecture rule.

For the architectural overview see `docs/ARCHITECTURE.md` § Multi-Tracker SDK. For the design rationale and the 14-section design doc see `docs/superpowers/specs/2026-04-30-sp3a-multi-tracker-sdk-foundation-design.md`.

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2. What changed (commit-level summary)

Phase commit counts

Phase	Scope	Commits
Phase 0 — Pre-implementation audit	Ledger seeding, fake-equivalence test scaffolding, fixture protocol	5
Phase 1 — Foundation	`core:tracker:api/registry/mirror/testing`, capability enum, DTOs	12
Phase 2 — RuTracker decoupling	git-mv from `core/network/rutracker`, `RuTrackerClient`, parser refit	40
Phase 3 — RuTor implementation	Descriptor, parsers, feature impls, fixtures, `RealRuTorIntegrationTest`	41
Phase 4 — Mirror health + UI	`MirrorHealthCheckWorker`, `CrossTrackerFallbackPolicy`, `feature:tracker_settings`	20
Phase 5 — Constitution + Challenges	Seventh Law, 8 Compose UI Challenge Tests, scripts/ci.sh, tag gate	26
Misc	Seventh Law landing, JVM-17 hardening, bluff audit wraps	4
Docs polish (this release)	README, ARCHITECTURE, SDK guide, per-module READMEs, release notes	5
SP-3a total		**153**

Test count delta (SP-3a → 1.2.0)

Surface	Count
New `*Test.kt` files in `core/tracker/*`	43
New `*Test.kt` files anywhere in the repo	60+
Compose UI Challenge Tests in `:app`	8
Real-tracker integration test (gated)	1 (`RealRuTorIntegrationTest`)

Pre-SP-3a the project had **one** unit-test file (`EndpointConverterTest`) and zero Compose UI tests. The 1.2.0 release ships **>60 unit-test files** including the eight Compose UI Challenge Tests that are the operator-attested acceptance gate.

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3. Latent findings summary

Tracked in `docs/superpowers/specs/2026-04-30-sp3a-coverage-exemptions.md`:

Finding	State	Mitigation trigger
LF-1	OPEN	First `MenuViewModel` test path that exercises bookmarks
LF-2	OPEN	Re-introducing any `healthcheck:` block on lava-api-go
LF-3	RESOLVED	(Phase 2 Section E wrap-up; Tracker-SDK pin enforces JVM 17)
LF-5	OPEN	`RuTrackerDescriptor` declares UPLOAD/ USER_PROFILE without backing feature interfaces — fix before any phase that depends on those capabilities
LF-6	OPEN	`TorrentItem.sizeBytes` permanently null for rutracker — fix before any cross-tracker numeric-size comparison

None of these findings block the 1.2.0 release. Each is a forward-looking tripwire with a documented mitigation trigger.

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4. Operator verification checklist

> **Tag is BLOCKED until every box below is ticked.** `scripts/tag.sh` mechanically refuses without the evidence pack.

Pre-conditions

- [ ] `git rev-parse HEAD` matches the commit being tagged.
- [ ] A physical Android device (API 26+, internet access) is connected; `adb devices` shows it as `device`.
- [ ] Host machine plugged in (not battery-only).
- [ ] `KEYSTORE_PASSWORD` and `KEYSTORE_ROOT_DIR` set in `.env`.
- [ ] `submodules/tracker_sdk/` is at the pinned hash; `git submodule status` shows no `-` or `+` prefix.

Build + install

```
./gradlew --no-daemon :app:assembleDebug
./gradlew --no-daemon :app:installDebug
```

- [ ] APK builds + installs without error.
- [ ] App icon `Lava (dev)` appears in launcher.
- [ ] First launch shows home screen without a crash dialog.

Local CI gate (Layer 2 of the pre-push hook, full mode)

```
./scripts/ci.sh --full
```

- [ ] Exit code 0.
- [ ] Output saved as `.lava-ci-evidence/Lava-Android-1.2.0-1020/ci.sh.json`.

8 Challenge Tests (C1–C8)

Each test is documented in `app/src/androidTest/kotlin/lava/app/challenges/Challenge0<n>*Test.kt`. For each `n` ∈ {1..8}:

1. Run / perform the scenario on the device.
2. Capture a screenshot of the user-visible state described in the test's primary assertion.
3. Apply the falsifiability mutation listed in the test header, re-run, capture verbatim failure assertion.
4. Revert mutation, re-run, confirm pass.
5. Update `.lava-ci-evidence/Lava-Android-1.2.0-1020/challenges/C<n>.json`:

```json
{
  "challenge_id": "C<n>",
  "status": "VERIFIED",
  "test_run_timestamp": "<iso8601>",
  "device": "<model + Android version>",
  "rehearsal_observed_failure": "<verbatim assertion message>",
  "screenshots_sha256": ["<hash1>", "<hash2>"]
}
```

Checklist:

- [ ] **C1** App launch + tracker selection — RuTracker + RuTor both visible in Settings → Trackers; tapping either updates active.
- [ ]

- [ ] **C2** Authenticated search on RuTracker — login, search "ubuntu", ≥1 result row with size + seeders.
- [ ] **C3** Anonymous search on RuTor — switch to RuTor, search "ubuntu", ≥1 result row with size + seeders. No login prompt.
- [ ] **C4** Switch tracker and re-search — banner changes from "Results from RuTracker" to "Results from RuTor".
- [ ] **C5** View topic detail — title, description, file list, magnet URI button all rendered.
- [ ] **C6** Download `.torrent` file — file written to app downloads dir, parses as bencoded torrent (`info.pieces` present).
- [ ] **C7** Cross-tracker fallback accept — with all RuTracker mirrors UNHEALTHY (test seed), modal appears, "Try RuTor" shows RuTor results.
- [ ] **C8** Cross-tracker fallback dismiss — same setup, "Cancel" shows explicit failure UI; **NO silent fallback to RuTor**.

Mirror smoke

- [ ] `submodules/tracker_sdk` at pinned hash; `scripts/sync-tracker-sdk-mirrors.sh --check` reports OK.
- [ ] Mirror smoke output saved to `.lava-ci-evidence/Lava-Android-1.2.0-1020/mirror-smoke/`.

Bluff hunt (Seventh Law clause 5)

```
./scripts/bluff-hunt.sh
```

- [ ] Pass — 5 randomly selected `*Test.kt` files mutate-and-fail.
- [ ] Output appended to `.lava-ci-evidence/bluff-hunt/<date>.json`.

Real-device attestation sign-off

- [ ] Replace `status: PENDING_OPERATOR` with `status: VERIFIED` at the top of `.lava-ci-evidence/Lava-Android-1.2.0-1020/real-device-verification.md`.
- [ ] Append the sign-off block:

```
verified_by:         <operator name or signer key id>
verified_at:         <iso8601 timestamp>
commit_sha:          <git rev-parse HEAD at verification time>
device:              <model + Android version + IMEI/serial>
evidence_zip_sha256: <sha256 of zipped evidence pack>
```

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5. Tag protocol (after operator attestation lands)

Once every box in §4 is ticked and `real-device-verification.md` reports `status: VERIFIED`:

```
# 1. Confirm pre-conditions one more time.
git rev-parse HEAD                              # matches the tagged commit
git submodule status                            # no -/+ prefix on Tracker-SDK
ls .lava-ci-evidence/Lava-Android-1.2.0-1020/challenges/ \
  | xargs -I{} grep -l '"status".*"VERIFIED"' \
    .lava-ci-evidence/Lava-Android-1.2.0-1020/challenges/{}
# expected: C1.json … C8.json all listed (eight matches)

# 2. Tag + push, all 4 upstreams.
./scripts/tag.sh --app android --bump patch

# scripts/tag.sh internally:
#   - calls require_evidence_for_android (validates the evidence pack)
#   - creates Lava-Android-1.2.0-1020 tag at HEAD
#   - pushes the tag to github + gitflic + gitlab + gitverse
#   - bumps versionName to 1.2.1 / versionCode to 1021 in :app
#   - commits the bump and pushes the bump commit
#   - per clause 6.C: verifies post-push tip-SHA convergence across
#     all four mirrors; refuses to report success on divergence

# 3. Confirm convergence.
for r in github gitflic gitlab gitverse; do
  echo "$r: $(git ls-remote $r refs/tags/Lava-Android-1.2.0-1020 | awk '{print $1}')"
done
# expected: identical SHA on all four lines.
```

If `scripts/tag.sh` exits non-zero on the evidence-pack gate, treat the output as the authoritative diagnosis — fix the gap (typically a missing `VERIFIED` status on a Challenge Test attestation) and re-run. Routine bypass (`--no-evidence-required` outside dry-run rehearsals) is itself a Seventh Law incident.

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6. Deferred to SP-3a-bridge

Items intentionally deferred (not regressions; they were never SP-3a's scope):

- **Third tracker.** The 7-step recipe is in `docs/sdk-developer-guide.md`; the first third-tracker module will exercise it end-to-end.
- **`UploadableTracker` + `ProfileTracker` feature interfaces.** RuTracker's `RuTrackerInnerApi` already has the underlying scrapers; LF-5 tracks the surfacing.
- **Numeric `TorrentItem.sizeBytes` for RuTracker.** LF-6 tracks the byte-count parser refit.
- **`connectedAndroidTest` runner wiring.** The 8 Challenge Tests run as manual operator scenarios in 1.2.0; the gradle task wiring lands in SP-3a-bridge.
- **Region-aware mirror health probes.** `mirrors.json` schema reserves the `region` field; routing logic that uses it is post-1.2.0.

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7. Cross-references

- Architecture overview: `docs/ARCHITECTURE.md`
- Developer guide for adding a tracker: `docs/sdk-developer-guide.md`
- Coverage exemption ledger: `docs/superpowers/specs/2026-04-30-sp3a-coverage-exemptions.md`
- Full design doc: `docs/superpowers/specs/2026-04-30-sp3a-multi-tracker-sdk-foundation-design.md`
- Constitutional law: root `CLAUDE.md` (Anti-Bluff Pact, Seventh Law, Local-Only CI/CD rule, Decoupled Reusable Architecture rule)
- Per-module READMEs:
  - `core/tracker/api/README.md`
  - `core/tracker/client/README.md`
  - `core/tracker/registry/README.md`
  - `core/tracker/mirror/README.md`
  - `core/tracker/rutracker/README.md`
  - `core/tracker/rutor/README.md`
  - `core/tracker/testing/README.md`
  - `feature/tracker_settings/README.md`

Last updated: 2026-05-01 (SP-3a docs polish; HEAD prior to tag).